Compilation of DSA Coding Problems

This document presents a comprehensive topicwise compilation of all coding problems from the Data Structures & Algorithms (BCSC 1006) Weekly Assessments. Each section details the core problem-solving pattern, difficulty level.

Topic 1: Mathematical Programming & Basic Logic

Mathematical and logic-building problems evaluate basic control structures, loops, digit manipulation, and modular arithmetic.

#	Problem Title	Difficulty
1	Even or Odd Number	Easy
2	Count Digits	Easy
3	Sum of Digits	Easy
4	Reverse Number / Integer	Easy / Medium
5	Palindrome Number	Easy
6	Prime Number Check	Easy
7	GCD / HCF of Two Numbers	Easy
8	Factorial of a Number	Easy
9	Multiplication Table	Easy
10	Number Pyramid Pattern	Easy / Medium

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Topic 2: Arrays & Sequential Traversal

Array traversal and basic manipulations form the backbone of memory management, indexing, and basic transformations.

#	Problem Title	Difficulty
11	Find Largest / Maximum Element	Easy
12	Find Second Largest Element	Easy
13	Check if Array is Sorted	Easy
14	Count Even and Odd Numbers	Easy
15	Reverse an Array	Easy
16	Left Rotate Array by One Position	Easy
17	Left Rotate Array by K Positions	Medium

Topic 3: Two Pointer Technique & In-Place Manipulation

The Two Pointer pattern optimizes brute-force pair search and space complexity by operating from opposite bounds or fast-slow indices.

#	Problem Title	Difficulty
18	Two Sum	Easy
19	Two Sum II - Input Array Is Sorted	Medium

#	Problem Title	Difficulty
20	Remove Duplicates from Sorted Array	Easy
21	Move All Zeroes to End	Easy
22	Container With Most Water	Medium
23	Merge Two Sorted Arrays	Easy
24	Shift All Negative Numbers to Left	Medium
25	Check if Subsequence	Easy

Topic 4: Searching & Sorting Algorithms

Covers linear search, binary search variants, lower/upper bounds, and comparison-based sorting techniques.

#	Problem Title	Difficulty
26	Linear Search	Easy
27	Binary Search	Easy
28	Search Insert Position	Easy
29	Bubble Sort	Easy
30	Selection Sort	Easy
31	Insertion Sort	Easy

Topic 5: Prefix Sum & Subarray Analysis

Precomputing prefix values allows O(1) range queries and enables efficient contiguous subarray processing.

#	Problem Title	Difficulty
32	Prefix Sum / Range Sum	Medium
33	Maximum Subarray Sum (Kadane's)	Medium

Topic 6: Frequency Counting & Hashing

Utilizing frequency arrays or HashMaps to solve counting, distinct element, and missing/duplicate identification problems in O(N) time.

#	Problem Title	Difficulty
34	Character Frequency Count	Easy
35	Frequencies of Limited Range Array Elements	Easy
36	Count Distinct Elements	Easy
37	Find All Numbers Appearing Twice / Disappeared	Easy / Medium
38	Find Missing Number (1 to N)	Easy

Topic 7: String Manipulation & Operations

Focuses on mutable vs immutable string operations, StringBuilder usage, ASCII conversions, and pattern recognition.

#	Problem Title	Difficulty
39	Reverse a String / Character Array	Easy
40	Check Valid Palindrome	Easy
41	Merge Two Strings Alternately	Easy
42	Reverse Vowels of a String	Easy / Medium
43	Reverse Every Word in a Sentence	Medium
44	Simulate String Compression	Medium
45	Valid Anagram Check	Easy
46	Longest Common Prefix	Easy
47	Remove Duplicate / Consecutive Duplicate Characters	Easy
48	Count Vowels and Consonants	Easy
49	Convert Lowercase to Uppercase (and vice versa)	Easy
50	Valid Parentheses	Easy / Medium

Topic 8: Fixed-Size Sliding Window

Fixed window technique optimizes $O(N \times K)$ brute force to $O(N)$ by maintaining window sums/counts when moving index by index.

#	Problem Title	Difficulty
51	Maximum Sum Subarray of Size K	Easy / Medium
52	Maximum Average Subarray I	Easy
53	Maximum Number of Vowels in Substring of Length K	Medium
54	Substrings of Size Three with Distinct Characters	Easy
55	Defuse the Bomb	Easy
56	K Radius Subarray Averages	Medium
57	Maximum Sum of Distinct Subarrays With Length K	Medium
58	Contains Duplicate II	Easy
59	Find All Anagrams in a String	Medium
60	Permutation in String	Medium
61	First Negative Integer in Every Window of Size K	Medium
62	Count Distinct Elements in Every Window	Medium

Topic 9: Variable-Size Sliding Window

Variable window pattern dynamically expands the right pointer to meet a condition and shrinks the left pointer to optimize window length.

#	Problem Title	Difficulty
63	Longest Substring Without Repeating Characters	Medium
64	Maximum Length Substring With Two Occurrences	Easy
65	Fruit Into Baskets	Medium
66	Max Consecutive Ones III	Medium
67	Minimum Size Subarray Sum	Medium
68	Subarray Product Less Than K	Medium

Topic 10: Binary Search

Binary Search algorithms operate on sorted arrays or monotonic search spaces to efficiently locate targets or optimal values in O(log N) time.

#	Problem Title	Difficulty
69	Koko Eating Bananas	Medium
70	Find First and Last Position of Element in Sorted Array	Medium